

Highlights

Optimization of a Recycling N-Stage Activated Sludge System with a Clarifier using A Physically Constrained Surrogate

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- One surrogate represents the complete recycling N -stage activated sludge system.
- A unique scaled- L_2 QP enforces mixer, reactor, and Clarifier physics.
- Ten Clarifier-layer solids states make the SRT inventory physically traceable.
- A full-rank scaled quadratic basis gives one fixed multiresponse fit.
- Mechanistic simulations separate prediction fidelity from decision quality.

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ABSTRACT

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